Table S1. Summary of model features estimated, fixed, or derived in the single-region estimation model (EM).

Model feature	**Treatment in EM**	**Notes**
Spatial structure	Fixed	Single-region; no movement or population components.
Mean recruitment	Estimated	Aggregate mean.
Recruitment deviation SD	Estimated	IID lognormal recruitment deviations.
Age-2+ deviation SD	Estimated	IID lognormal age-2+ deviations.
Initial numbers-at-age	Equilibrium initialization	WHAM equilibrium approach.
Annual fishing mortality by fleet	Estimated	Commercial and recreational fleets.
Commercial selectivity	Partly estimated	Ages 1-3 estimated; ages 4-8+ fixed at 1.
Recreational selectivity	Partly estimated	Ages 1-4 estimated; ages 5-8+ fixed at 1.
Survey selectivity	Partly estimated	Age 1 estimated; ages 2-8+ fixed at 1.
Survey catchability	Estimated	Aggregate survey index.
Catch observation error CVs	Fixed	Commercial = 0.05; recreational = 0.15.
Survey observation error CV	Fixed	CV = 0.4.
Age-composition ESS	Fixed	ESS = 50.
Natural mortality	Fixed	M = 0.4 for all ages.
Weight-at-age	Fixed	Aggregated before fitting EM.
Maturity-at-age	Fixed	Aggregated before fitting EM.
F40% reference point	Derived	Calculated from each EM fit.